

Project

Industrial Sensor Enclosure Design

1. Objective

Define the mechanical design requirements for an industrial-grade sensor enclosure capable of operating in harsh manufacturing environments.

2. Design Requirements

- IP67 protection
- Corrosion-resistant housing
- Shock-resistant enclosure
- Modular assembly
- Easy maintenance access
- Lightweight construction

3. Materials

Component	Material
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Outer Housing	Aluminum Alloy 6061
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Internal Frame	Stainless Steel 316
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Seal	Silicone Rubber
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Fasteners	Stainless Steel
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4. Mechanical Specifications

Parameter	Value
Weight	<2 kg
Height	220 mm
Width	160 mm

Parameter	Value
Depth	110 mm
Operating Pressure	2 bar

5. Environmental Conditions

- Operating Temperature: -20°C to 70°C
 - Storage Temperature: -40°C to 85°C
 - Humidity: 5–95%
 - UV Resistant
 - Dustproof
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6. Acceptance Tests

- Drop Test
- Vibration Test
- Shock Test
- Water Ingress Test
- Dust Test
- Thermal Expansion Test